

Tara Finance Agent

Overview:

Tara Finance Agent is a financial analytics assistant built using Mastra, PostgreSQL, and TypeScript.

The agent provides insights across both spending and investments through natural language conversations. Users can ask questions about their spending habits, merchants, categories, trends, portfolio performance, fund rankings, and holdings.

The solution combines structured financial data with LLM-powered reasoning and tool calling to deliver accurate financial insights.

Features

Spending Analytics

- Total spending analysis
- Category-wise spending
- Merchant-wise spending
- Top merchants
- Category distribution
- Spending trends
- Largest transactions
- Spending comparisons between periods

Example Questions

- How much did I spend on Groceries?
 - Show my top merchants.
 - What are my biggest transactions?
 - Compare my spending in January vs February.
 - Show my spending trend.
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Portfolio Analytics

- Portfolio holdings
- Portfolio value
- Largest holding
- Best performing fund
- Worst performing fund
- Fund ranking
- Holding returns

Example Questions

- What is my largest holding?
 - Which fund is performing best?
 - Which fund is performing worst?
 - Rank all my funds.
 - Show holding returns.
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Technology Stack

- Mastra
- TypeScript
- Node.js
- PostgreSQL
- Zod

Database Design

The solution uses PostgreSQL for structured financial data storage.

Tables

Transactions

Stores all spending transactions.

Column	Type
id	TEXT
date	DATE
merchant	TEXT
category	TEXT
amount	NUMERIC
currency	TEXT
memo	TEXT

Funds

Stores mutual fund metadata.

Column	Type
id	TEXT
name	TEXT
category	TEXT

Fund_nav

Stores historical NAV values.

Column	Type
id	SERIAL
fund_id	TEXT
nav_date	DATE
nav	NUMERIC

Holdings

Stores user holdings.

Column	Type
id	SERIAL
fund_id	TEXT
units	NUMERIC
purchase_date	DATE
purchase_nav	NUMERIC

Data Ingestion

Financial data is loaded into PostgreSQL through an ingestion script.

Supported datasets:

```
Bash
./data/sample_a
```

Run:

```
Bash
DATA_DIR=./data/sample_a npx tsx scripts/ingest.ts
```

Example output:

```
Plain text
Inserted 1500 transactions
Inserted 8 funds
Inserted 192 NAV records
Inserted 8 holdings
Ingestion complete!
```

The ingestion script loads:

- transactions.json
- funds.json
- holdings.json

into PostgreSQL tables.

Installation

Clone Repository

```
Bash
git clone <repository-url>
cd tara-finance-agent
```

Install Dependencies

```
Bash
npm install
```

Environment Variables

Create a .env file:

```
env
DATABASE_URL=your_postgres_connection_string
```

Running the Application

Start Mastra:

```
Bash
npm run dev
```

Mastra Studio:

```
Plain text
http://localhost:4111
```

Agent Architecture

The solution consists of:

Tara Finance Agent

Primary conversational agent responsible for:

- * Understanding user intent
- * Selecting appropriate tools
- * Formatting financial responses
- * Generating concise insights

Spending Tool

Handles:

- * Category spending
- * Merchant spending
- * Spending summaries
- * Spending trends
- * Spending comparisons
- * Largest transactions
- * Category distribution

Portfolio Tool

Handles:

- * Fund performance
- * Portfolio valuation
- * Fund rankings
- * Largest holdings
- * Holding returns

Example Queries

Spending

Plain text

How much did I spend on Groceries?

Plain text

Show my top merchants.

Plain text

What are my largest transactions?

Plain text

Compare my spending in January vs February.

Portfolio

Plain text

What is my largest holding?

Plain text

Which fund is performing best?

Plain text

Rank all my funds.

Plain text

Show holding returns.

Project Structure:

Plain text

```
src/  
├── lib/  
│   └── db.ts  
  
├── mastra/  
│   ├── agents/  
│   ├── tools/  
│   ├── scorers/  
│   └── index.ts  
  
scripts/  
└── ingest.ts  
  
data/  
└── sample_a
```

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